

Literature-implied answer for arXiv:1605.00237

All massive integer-spin field strengths have the conjectured fine ultraviolet extension

Status. literature_implied_answer (full conjectured scope). This is not a new theorem produced by the run. It is an agent-identified implication from a later all-spin lemma, combined with the extension criterion proved and used in the source paper.

Source conjecture. Joseph C. Várilly and José M. Gracia-Bondía, *Stora's fine notion of divergent amplitudes*, arXiv:1605.00237, Section 6, PDF page 9, prove the massive spin-one and spin-two cases and then state:

“We conjecture that our conclusions extend to all the massive $F_{\mu_1\nu_1,\dots,\mu_h\nu_h}$ -fields.”

Here the conclusion is that the off-diagonal time-ordered two-point function has the optimal fine ultraviolet behavior and extends to the diagonal without further renormalization.

Decisive later theorem. Jens Mund and Erichardson T. de Oliveira, *String-localized free vector and tensor potentials for massive particles with any spin: I. Bosons*, arXiv:1609.01667, Lemma 14, PDF page 28, construct the massive spin- s point-local field strength and prove that its on-shell two-point function is a homogeneous polynomial $P_s(p)$ of degree $2s$. Their proof writes the all-spin intertwiner as the symmetric trace-free projection of s spin-one field-strength intertwiners; each scalar-product term is consequently a product of s homogeneous quadratic spin-one factors.

Why this answers the conjecture. Off the diagonal the corresponding Feynman two-point function is represented by $P_s(\partial)D_m^F$. Its leading massless singularity has the form

$$P_s(\partial)\frac{1}{x^2-i0}=\frac{H_{2s}(x)}{(x^2-i0)^{2s+1}},$$

where H_{2s} is homogeneous of degree $2s$. It is harmonic: away from the origin the left side is killed by $\square$, and the radial identity in Section 3 of the source paper, with $\alpha=-(2s+1)$ and $k=2s$, then forces $\square H_{2s}=0$.

The source paper's NST criterion compares the harmonicity degree k with the degree of divergence $2r-k-4$. Here $k=2s$ and $r=2s+1$, so

$$2s > 2(2s+1) - 2s - 4 = 2s - 2.$$

Hence the leading term has a unique Lorentz-covariant homogeneous extension. The massive Riesz/associate-homogeneity construction in Section 6 of the source paper canonically extends the lower mass-dependent terms. Thus Mund–de Oliveira's Lemma 14 provides precisely the all-spin algebraic input missing from the two cases checked in the source.

An on-shell polynomial determines the off-diagonal Feynman numerator up to a multiple of p^2-m^2 ; after division by the Klein–Gordon denominator, that freedom is a contact term supported on the diagonal. It therefore does not create an obstruction to the asserted extension.

Literature-implied conclusion. For every positive integer s , the massive point-local field strength $F_{\mu_1\nu_1,\dots,\mu_s\nu_s}$ has the fine ultraviolet extension to the diagonal conjectured in arXiv:1605.00237.

Provenance and scope. The supporting paper cites arXiv:1605.00237 when discussing massive associate-homogeneity, but it does not announce Lemma 14 as a resolution of the displayed conjecture. The relation is therefore an agent-identified literature implication, not an explicit literature answer. It covers every positive integer spin in the conjectured family; it makes no claim about interacting composite fields or higher time-ordered products.

Search evidence. The run’s cheap indexes had no prior result for arXiv:1605.00237. A bounded source and citation search used the arXiv id, the displayed field-strength notation, “all massive”, “homogeneous polynomial of degree $2s$ ”, and “associate homogeneity”. It located arXiv:1609.01667 and the cross-citation above, but no paper explicitly claiming to resolve the exact conjecture.

References

- [1] J. C. Várilly and J. M. Gracia-Bondía, *Stora’s fine notion of divergent amplitudes*, arXiv:1605.00237; Nuclear Physics B 912 (2016), 28–37.
- [2] J. Mund and E. T. de Oliveira, *String-localized free vector and tensor potentials for massive particles with any spin: I. Bosons*, arXiv:1609.01667; Communications in Mathematical Physics 355 (2017), 1243–1280.